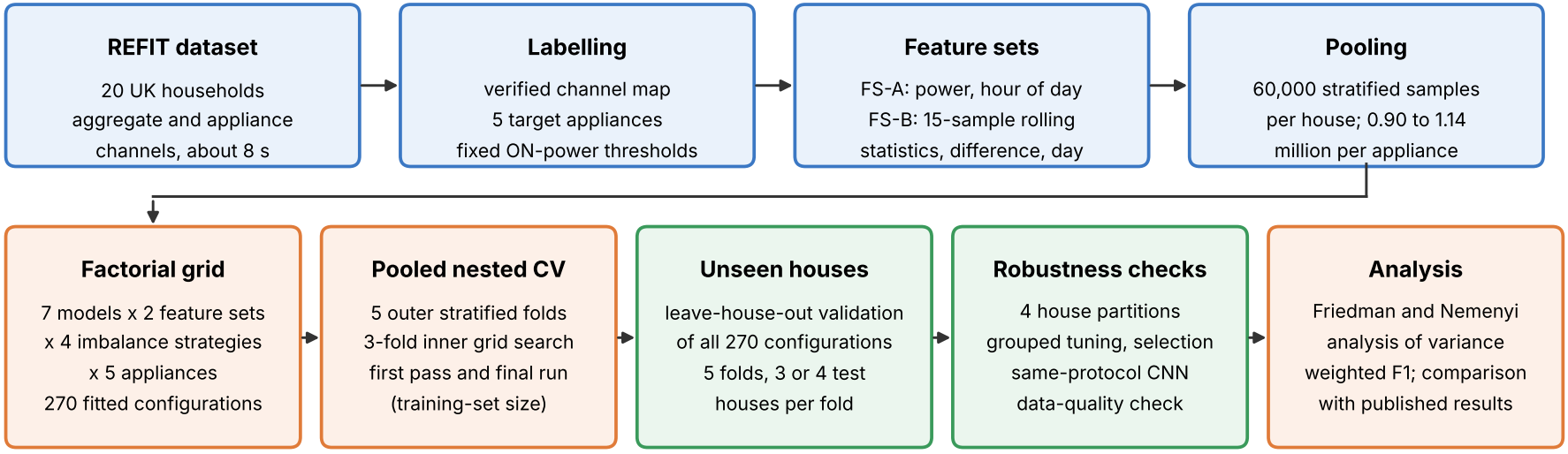
REFIT dataset
20 UK households
aggregate and appliance
channels, about 8 s
Labelling
verified channel map
5 target appliances
fixed ON-power thresholds
Feature sets
FS-A: power, hour of day
FS-B: 15-sample rolling
statistics, difference, day
Pooling
60,000 stratified samples
per house; 0.90 to 1.14
million per appliance
Factorial grid
7 models x 2 feature sets
x 4 imbalance strategies
x 5 appliances
270 fitted configurations
Pooled nested CV
5 outer stratified folds
3-fold inner grid search
first pass and final run
(training-set size)
Unseen houses
leave-house-out validation
of all 270 configurations
5 folds, 3 or 4 test
houses per fold
Robustness checks
4 house partitions
grouped tuning, selection
same-protocol CNN
data-quality check
Analysis
Friedman and Nemenyi
analysis of variance
weighted F1; comparison
with published results